Materials Science & Engineering A 959 (2026) 150047

Contents lists available at ScienceDirect

# Materials Science & Engineering A

journal homepage: www.elsevier.com/locate/msea

# Effects of Hf and Re on the microstructure and mechanical properties of $Nb_{1.6}Mo_{0.3}C_{0.04}$ alloy

Y.J. Hou [a], J.X. Fang [a,*], Q. Su [b], H.T. He [a], X. Mei [a], G.F. Long [a], L.Y. Hao [a], M. Wen [c,**]

[a] Faculty of Mechanical and Electrical Engineering, Kunming University of Science and Technology, Kunming, 650500, China
[b] CNPC Offshore Engineering Company Limited, Beijing, 100176, China
[c] State Key Laboratory of Advanced Technologies for Comprehensive Utilization of Platinum Metals, Kunming, 650106, China

ARTICLE INFO

*Keywords:*
Nb–Mo-based refractory alloys
Carbide precipitation
High-temperature strength
Room-temperature ductility

ABSTRACT

Carbide-strengthened Nb–Mo-based refractory alloys with nominal compositions of $Nb_{1.6}Mo_{0.3}C_{0.04}$, $Nb_{1.6}Mo_{0.3}C_{0.04}Hf_{0.04}$, $Nb_{1.6}Mo_{0.3}C_{0.04}Re_{0.04}$, and $Nb_{1.6}Mo_{0.3}C_{0.04}Hf_{0.04}Re_{0.04}$ were fabricated to elucidate the roles of Hf and Re in microstructure evolution and compressive properties. All alloys consisted of a BCC matrix as the predominant phase, containing dispersed, semi-coherent platelet-like FCC carbide precipitates. Re addition suppressed carbide precipitation, whereas Hf promoted carbide precipitation and counteracted the Re-induced suppression; moreover, Hf reduced the lattice misfit at the FCC/BCC interface. The $Nb_{1.6}Mo_{0.3}C_{0.04}$ alloy did not fracture under room-temperature compression, while surface cracks appeared when the compressive strain exceeded 50%, giving a yield strength of 604.29 MPa. Minor additions of Re, Hf, or combined Re and Hf additions increased the room-temperature yield strength effectively, with a maximum increment of 37.4%; However, the Re-only alloy exhibited degraded compressive ductility, showing pronounced surface cracking once the strain exceeded 30%. In contrast to room-temperature mechanical properties, the addition of elements such as Re and Hf exerted a more pronounced effect on the high-temperature strength of the alloy. Compared to the base composition $Nb_{1.6}Mo_{0.3}C_{0.04}$, the separate addition of Re, the separate addition of Hf, and the combined addition of Re + Hf increased the yield strength at 1450 °C by 41.5%, 39.5%, and 91.7 %, respectively, reaching 256.45 MPa, 252.80 MPa, and 347.43 MPa. The $Nb_{1.6}Mo_{0.3}C_{0.04}Re_{0.04}Hf_{0.04}$ alloy achieved a specific yield strength of 38.14 MPa $cm^3/g$ at 1450 °C, which is 1.19 times that of the NbMoTaW alloy at 1400 °C, indicating outstanding high-temperature strength and excellent resistance to high-temperature softening. The results demonstrate that small additions of Hf and Re can significantly tailor carbide precipitation (fraction, intrinsic characteristics, and matrix/precipitate misfit), enabling substantial strengthening at elevated temperatures without compromising room-temperature compressive ductility. These alloys are thus promising candidates for ultrahigh-temperature structural components.

## 1. Introduction

The high-temperature performance of ultrahigh-temperature structural materials is a key factor governing technological advances in aerospace, nuclear energy, and related fields, yet existing materials remain unable to meet the demanding service requirements.

For instance, at temperatures above approximately 1200 °C, Ni-based superalloys generally suffer from a pronounced loss of microstructural stability, accompanied by a substantial reduction in mechanical strength [1–3]. In the case of the Haynes 230 superalloy, the yield strength decreases to approximately 15 MPa at 1200 °C [4], which limits its applicability for load-bearing hot-section components, such as turbine blades.

Conventional refractory alloy systems such as W–Mo, Ta–W, and W–Re suffer from high density, low specific strength, and pronounced room-temperature brittleness. Developing ultrahigh-temperature structural materials with a balanced combination of properties has therefore become a critical challenge. Nb–Mo-based alloys exhibit service temperature capabilities well beyond those of Ni-based superalloys, together with high specific strength. They have been utilized in aerospace propulsion systems, including rocket-engine thrust chambers and nozzles, and are regarded as potential materials for hot-section

* Corresponding author.
** Corresponding author.
*E-mail addresses:* fangjx6@gmail.com (J.X. Fang), wen@ipm.com.cn (M. Wen).

https://doi.org/10.1016/j.msea.2026.150047
Received 29 December 2025; Received in revised form 11 February 2026; Accepted 3 March 2026
Available online 4 March 2026
0921-5093/

components in next-generation gas turbines and aero-engines [5–7]. Nevertheless, further engineering implementation requires the concurrent enhancement of high-temperature strength and room-temperature ductility, so as to satisfy both high-temperature load-bearing performance and processing-related manufacturability requirements.

In recent years, extensive efforts have been devoted to enhancing the high-temperature load-bearing capability of Nb-based refractory alloys. However, existing commercial Nb-based alloys still face a critical bottleneck, namely insufficient strength above 1000 °C. For example, the typical solid-solution-strengthened Nb alloy C-103 (Nb–10Hf–1Ti, wt.%) exhibits a relatively low yield strength (<150 MPa) above 1000 °C. Another commercial Nb alloy, C-3009 (Nb–20Hf–6W, wt.%), demonstrates a markedly improved high-temperature strength, with a yield strength reaching approximately 397 MPa at 1000 °C; nevertheless, this improvement is accompanied by a higher density and increased processing difficulty [8,9]. Consequently, strengthening strategies are still required for Nb–Mo-based alloy systems to achieve higher high-temperature strength while maintaining engineering applicability.

To enhance the high-temperature strength of Nb–Mo-based alloys, research efforts have mainly followed two major approaches. The first approach involves alloying and high-entropy design, in which solid-solution strengthening is achieved by introducing pronounced lattice distortion and elastic modulus mismatch to improve high-temperature load-bearing capability. However, a prominent contradiction has been widely observed: improvements in high-temperature strength are often accompanied by a severe increase in intrinsic brittleness. For example, $(HfCo)_{100-x}(NbMo)_x$ alloys with a dual-phase BCC–$B_2$ structure exhibit high yield strength at elevated temperatures, whereas their room-temperature compressive fracture strain decreases to approximately 5% when the composition is optimized for strength [10]. Similarly, multi-principal-element Mo–Nb–Hf–Zr–Ti alloys show strong composition-dependent strengthening behavior, yet the room-temperature fracture strain of the strongest compositions remains below ~12% [11]. Typical refractory high-entropy alloys such as NbMoTaW possess high yield strength at 1400 °C but exhibit extremely poor room-temperature ductility [4]. Further attempts to optimize the strength–ductility balance through Ti alloying of NbMoTaW, either in the as-cast state or fabricated by laser powder bed fusion (LPBF), can improve strength in certain temperature ranges; however, the room-temperature fracture strain generally remains below ~8–10%, indicating that optimization strategies dominated by solid-solution strengthening alone are insufficient to effectively overcome the strength–ductility trade-off [12,13]. The second approach involves the introduction of second phases to achieve dispersion strengthening. For instance, Nb–Mo-matrix composites reinforced with $ZrB_2$ or $ZrB_2$–SiC particles exhibit significantly enhanced room-temperature strength, yet their room-temperature compressive fracture strain is typically lower than ~15%, suggesting that stress concentration and interfacial incompatibility induced by the reinforcement may dominate the failure process [14–16]. Similarly, in-situ carbide-containing $Mo_{0.5}NbHf_{0.5}ZrTiC_x$ alloys show pronounced brittleness, reflecting the limited plastic compatibility generally associated with hard second-phase strengthening [17]. Even though metal–carbide eutectic designs can achieve exceptionally high high-temperature strength, increasing carbide content often leads to a substantial degradation in room-temperature ductility, further highlighting the universal nature of the “high-temperature strength–room-temperature plasticity” trade-off [18]. In summary, existing studies indicate that, within Nb–Mo-based alloy systems, achieving a substantial enhancement in high-temperature strength while simultaneously retaining good room-temperature ductility remains highly challenging. This challenge underscores the urgent need for alternative microstructural design strategies that can deliver robust high-temperature strengthening without inducing pronounced room-temperature ductility degradation.

The authors’ previous studies revealed that two types of nanoscale carbides exhibit the potential for plastic deformation at room temperature. Specifically, certain FCC-structured MC-type carbides can accommodate room-temperature plastic deformation through stacking-fault formation on the (111) plane, whereas some close-packed hexagonal $M_2C$-type carbides can deform plastically via basal slip on the (0001) plane. These two types of carbides can significantly enhance high-temperature strength while retaining a degree of coordinated plastic deformation at room temperature [19,20]. Owing to the high similarity between Nb–Mo and Ta–W alloy systems in terms of crystal structure and carbide formation behavior, the strengthening strategy based on room-temperature deformable carbides is expected to be directly transferable to Nb–Mo-based alloys.

In mature alloy systems, Hf has been demonstrated to exhibit a strong chemical affinity for carbon, promoting the formation of stable MC-type carbides and thereby ensuring high-temperature strength. In both Ni-based and Nb-based alloys, Hf effectively regulates the stability, volume fraction, and morphology of carbides, consequently influencing strength and ductility [21,22]. In the authors’ previous studies on BCC-matrix refractory alloys $Re_{0.1}Hf_xTa_{1.6}W_{0.4}(TaC)_y$, Hf and C were confirmed to form MC-type carbides in a limited manner, while simultaneously tailoring grain size and carbide dimensions, which markedly enhanced the high-temperature load-bearing capability of the alloys [19,23]. Meanwhile, Re is widely recognized as a key high-temperature strengthening element in Ni-based superalloys, with its contribution closely associated with its low diffusion coefficient, high melting point, and partitioning behavior in the matrix [24]. In refractory alloy systems, available evidence further indicates that minor Re alloying can enhance strength while retaining good room-temperature ductility. Accordingly, Re can be regarded as an alloying element capable of providing pronounced high-temperature strengthening without a severe penalty in room-temperature ductility [25–27]. Previous studies on Ta–W-based refractory alloys by the authors also demonstrated that Re addition increases the recrystallization temperature, suppresses premature high-temperature softening, and enhances strength through grain refinement and solid-solution strengthening [20,28].

Based on the established roles of Hf in regulating carbide precipitation and Re in providing high-temperature strengthening, and in conjunction with the authors’ prior studies on Nb–Mo-based alloys [29], the present work systematically investigates the effects of Hf and Re on carbide precipitation behavior, microstructural evolution, and mechanical properties, with the aim of achieving enhanced high-temperature strength without pronounced degradation of room-temperature ductility.

In this work, four carbide-strengthened Nb–Mo-based alloys, namely $Nb_{1.6}Mo_{0.3}C_{0.04}$, $Nb_{1.6}Mo_{0.3}C_{0.04}Hf_{0.04}$, $Nb_{1.6}Mo_{0.3}C_{0.04}Re_{0.04}$, and $Nb_{1.6}Mo_{0.3}C_{0.04}Hf_{0.04}Re_{0.04}$, were prepared by arc melting. Their microstructures as well as room-temperature and high-temperature mechanical properties were characterized. The mechanical performance was also compared with that of other representative refractory alloys. The roles of carbides in strengthening, and the effects of Re and Hf on microstructure and mechanical behavior, were analyzed and discussed. This work provides a reference for developing a new generation of high-performance Nb–Mo-based alloys.

## 2. Material and methods

### 2.1. Sample preparation

Samples were prepared using a DHL1250 vacuum arc-melting furnace (SKY Technology Development Co., Ltd., China). Four alloys with nominal compositions of $Nb_{1.6}Mo_{0.3}C_{0.04}$ (referred to as #1), $Nb_{1.6}Mo_{0.3}C_{0.04}Hf_{0.04}$ (referred to as #2-Hf), $Nb_{1.6}Mo_{0.3}C_{0.04}Re_{0.04}$ (referred to as #3-Re), and $Nb_{1.6}Mo_{0.3}C_{0.04}Hf_{0.04}Re_{0.04}$ (referred to as #4-HfRe) were fabricated, where the subscripts denote atomic ratios (The elemental concentrations adopted for the four alloys were selected based on previously reported works and the authors’ prior works [19,20,22,23,25,28]). High-purity elemental metals with purities higher than

99.95 wt% were used as starting materials. For #1 and #3-Re, carbon was introduced via NbC powder (5 μm, 99.9 wt%). For #2-Hf and #4-HfRe, both Hf and C were introduced via HfC powder (1–3 μm, 99.9 wt%).

Prior to melting, the carbide powder was wrapped with Nb foil and placed at the bottom of the copper crucible to prevent powder spattering during melting; the remaining elemental metals were placed on top of the wrapped carbide compact. The chamber was evacuated to $5 \times 10^{-4}$ Pa before backfilling with argon as a protective atmosphere. Arc melting was carried out under a high-purity Ar atmosphere at a current of 400–450 A. To improve chemical homogeneity within the ingots, electromagnetic stirring was applied during melting, and each ingot was remelted five times, with the button ingot flipped by 180° between successive melts. A schematic illustration of the melting setup is shown in Fig. S1. The as-melted products were button-shaped cast ingots, from which specimens were sectioned for subsequent characterization and mechanical testing.

### 2.2. Characterization of microstructure

Phase identification was performed by X-ray diffraction (XRD, D8, Bruker, Germany) using Cu Kα radiation. XRD patterns were collected over a 2θ range of 30 °–120° at a scanning rate of 2°/min. Prior to XRD measurements, the specimen surfaces were mechanically polished. The carbide fraction was quantified by Rietveld full-pattern refinement using the GSAS software.

Microstructural observations were carried out using a scanning electron microscope (SEM, SEM5000, GY Quantum, China). SEM specimens were prepared by a standard mechanical grinding and polishing procedure, followed by chemical etching. The etchant consisted of $HNO_3$:HF:$H_2O$ = 1:2:2 (vol. ratio), and the etching time was 10 s.

A transmission electron microscope (TEM, Talos F200X, USA) was employed to characterize the composition, structure, and plastic-deformation mechanisms of the precipitates. TEM foils were sectioned by wire electrical discharge machining (wire-EDM) into ~500 μm-thick slices, mechanically ground to ~50 μm, and finally thinned by ion milling to electron transparency.

### 2.3. Characterization of mechanical properties

Vickers microhardness was measured using a microhardness tester (HVS-1000AT, Hengyi Testing Instruments Co., Ltd., China). Prior to indentation, the specimen surfaces were prepared by mechanical grinding and polishing. For each alloy, at least five indentations were made at different locations, and the reported hardness value was taken as the average of the five measurements.

The actual density of the alloys was measured using an FA2204N electronic density analyzer (based on the Archimedes drainage method). Before testing, the specimen surfaces were prepared by mechanical grinding and polishing. For each alloy specimen, two independent measurements were taken, and the reported density value is the average of the two results.

Room-temperature and high-temperature compression tests were conducted using cylindrical specimens with the same geometry (5 mm in diameter and 7.5 mm in length). All specimens were mechanically ground prior to testing. Room-temperature compression was performed on a universal testing machine (UTM5105X, SUNS Technology Co., Ltd., China) at an initial strain rate of 0.001 $s^{-1}$.

High-temperature compression tests were carried out using a Gleeble 3500 thermomechanical simulator. Specimens were heated to 1450 °C at a rate of 10 °C/s and held for 5 min prior to compression. The compression tests were conducted at a strain rate of 0.01 $s^{-1}$ and were terminated when the compressive strain reached 50% or when fracture occurred.

## 3. Results and analysis

### 3.1. Phase composition analysis and microstructure

Fig. 1 shows the XRD patterns of the four alloys prepared in this work. As can be seen from the figure, the primary phase in all four alloys was identified as the BCC phase, with a minor amount of FCC phase also present. The lattice parameters of the BCC and FCC phases for these alloys are summarized in Table 1.

Fig. 2 shows the SEM micrographs of the four refractory alloys. Based on the X-ray diffraction results and previous reports [19,20,23,28], the platelet-like precipitates observed in all four alloys can be preliminarily identified as FCC-structured MC-type carbides. Pronounced differences are observed among the alloys in terms of grain size as well as the size and spatial distribution of the precipitates. As shown in Fig. 2 and Fig. S2, the #1 alloy exhibits the coarsest grains, with an average grain size of 298.33 μm and an average precipitate length of 27.23 μm. The platelet-like carbides are homogeneously dispersed in the matrix. With the addition of Re alone, the #3-Re alloy shows significant grain refinement to an average size of 178.82 μm, while the precipitates remain uniformly distributed with a comparable average length of 22.34 μm.

In contrast, the addition of Hf alone leads to pronounced microstructural refinement, with the average grain size reduced to 127.570 μm and the precipitate length markedly decreased to 2.940 μm. In this case, the carbides exhibit a mixed distribution characterized by the coexistence of clustered and dispersed platelet-like precipitates. When Hf and Re are added simultaneously (#4-HfRe), both the grain size (130.130 μm) and precipitate length (3.585 μm) are comparable to those of the Hf-containing alloy, and a similar mixed distribution of carbides is observed. (The grain size and precipitate dimensions were statistically determined from multiple SEM and optical micrographs using Nano Measure software.)

To clarify the crystal structure of the platelet-like precipitates and their crystallographic relationship with the matrix, transmission electron microscopy (TEM) was analyzed. Fig. 3(a) shows a TEM bright-field (BF) image of the #4-HfRe alloy together with the corresponding EDS elemental maps of the same area. As shown, the precipitates are enriched in Hf and C, contain a certain amount of Nb and a minor amount of Mo, but are depleted in Re. Fig. 3(b) presents the high-resolution TEM (HRTEM) image of Region 1 in Fig. 3(a) and the corresponding fast Fourier transform (FFT) pattern, confirming that the precipitates are FCC-structured carbides. In addition, the precipitates are semi-coherent with the BCC matrix. The following orientation relationship is observed: $[110]_{FCC}//[100]_{BCC}$; $(\bar{1}11)_{FCC}//(011)_{BCC}$.

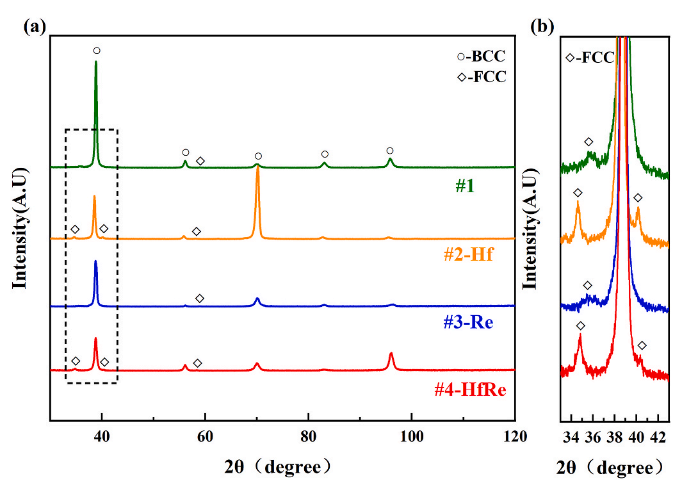


**Fig. 1.** (a) XRD patterns of the four alloys; (b) Enlarged view of the area marked by the dashed rectangle in (a). 2-Hf adapted from Ref. [29].

**Table 1**
Lattice parameters of different phases in the four alloys.

| Alloys | BCC (Å) | FCC (Å) |
|---|---|---|
| #1 | 3.2820 | 4.3841 |
| #2-Hf | 3.2920 | 4.4843 |
| #3-Re | 3.2851 | 4.3896 |
| #4-HfRe | 3.2822 | 4.4634 |

Based on Refs. [30,31], the orientation relationship can be identified as the Baker–Nutting orientation relationship, i.e., $[110]_{FCC}//[100]_{BCC}$; $(001)_{FCC}//(001)_{BCC}$.

### 3.2. Mechanical properties

Fig. 4 and Fig. S7 show the room-temperature compressive engineering stress–strain curves and room-temperature true stress–strain curves of the four alloys prepared in this work, respectively. The #1 alloy exhibits a yield strength of 604.29 MPa. With minor additions of Re, Hf, and combined Re and Hf additions, the room-temperature yield strength increases by 5.14%, 20.23%, and 37.43%, reaching 635.36 MPa, 726.54 MPa, and 830.48 MPa, respectively. In addition, the alloy containing only Re shows reduced room-temperature compressive ductility, with a fracture strain of 31.00%. In contrast, no obvious change in compressive ductility is observed for the Hf-containing alloy and the alloy containing both Hf and Re.

Fig. S3 shows the room-temperature hardness of the four alloys. The average Vickers hardness values of #1, #2-Hf, #3-Re, and #4-HfRe are 299.40 HV, 316.60 HV, 338.80 HV, and 368.60 HV, respectively. The hardness exhibits a positive correlation with the room-temperature compressive yield strength; namely, the Vickers hardness increases with increasing yield strength.

Fig. 5 and Fig. S8 show the high-temperature compressive engineering stress–strain curves at 1450 °C and the room-temperature true stress-strain curves of the four alloys prepared in this work, respectively. (The high-temperature compression tests were performed on single specimens; although compressive data are generally more repeatable than tensile data for defect-free metals, this remains a limitation of the present work [29,32,33].) The #1 alloy has a yield strength of 181.23 MPa and an ultimate compressive strength of 381.60 MPa. Compared with the room-temperature behavior, the additions of Re and Hf have a more pronounced effect on high-temperature strength. The yield strength at 1450 °C increases by 41.50%, 39.50%, and 91.70% for the Re-, Hf-, and Re + Hf - containing alloys, reaching 256.45 MPa, 252.80 MPa, and 347.43 MPa, respectively. Moreover, all four alloys exhibit good resistance to high-temperature softening. The compressive properties at room temperature and 1450 °C are summarized in Table 2 and Fig. 6.

Fig. S3 Fig. S3 shows photographs of the specimens after room-temperature compression. As observed, irregular surface cracks are present on all compressed specimens. The #1, #2-Hf, and #4-HfRe alloys did not fracture, whereas the #3-Re alloy fractured and exhibited minor spallation. As shown in Fig. S5, no open fracture surfaces suitable for SEM observation were formed after compression at 1450 °C; therefore, fracture morphology comparisons at high temperature could not be performed. Fig. S4 presents the positions and orientations for fracture surface characterization of the four alloys after room-temperature compression, and Fig. 7 shows high-magnification SEM images of the fracture morphologies of the four alloys after room-temperature compression. All alloys exhibit quasi-cleavage fracture characteristics, such as cleavage steps, together with locally observed plastic deformation features, including tearing ridges. This indicates that room-temperature fracture is predominantly brittle in nature, accompanied by a limited degree of plastic energy dissipation.

Notably, differences in fracture details are evident among the alloys. In the #1, #2-Hf, and #4-HfRe alloys, more pronounced local plastic deformation features are observed in addition to cleavage characteristics. In particular, the #2-Hf alloy exhibits shallow dimple-like features in certain regions, suggesting a higher degree of plastic involvement during fracture. In contrast, the #3-Re alloy remains dominated by quasi-cleavage features but shows more pronounced secondary cracking and local spallation; consistent with this, more severe macroscopic cracking and slight material detachment are observed after compression (Fig. S2), leading to a relatively lower fracture strain.

Although the four alloys exhibit similar fracture modes, differences in the extent of plastic deformation participation as well as crack propagation and spallation behavior are evident, which are consistent with the observed variations in room-temperature compressive fracture strain (Fig. 4).

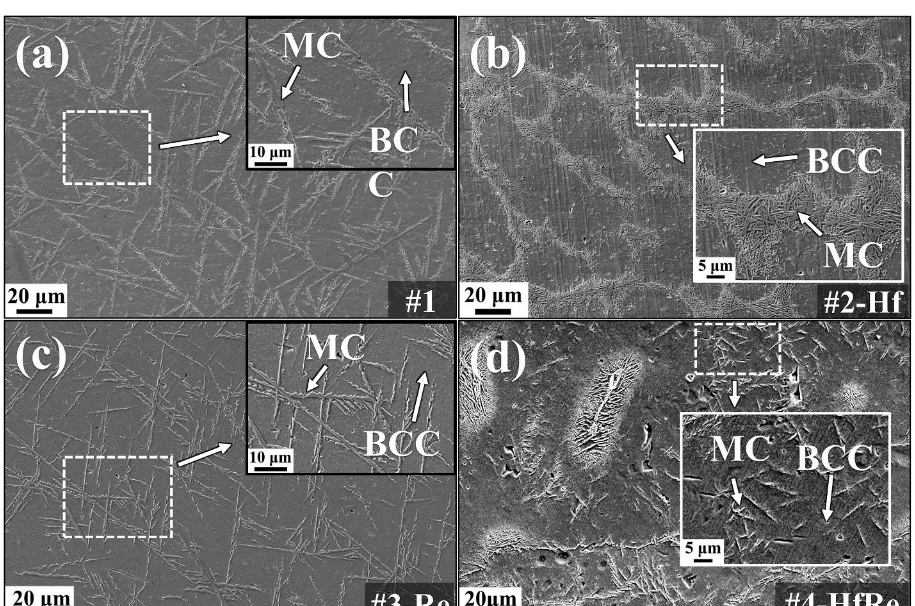


**Fig. 2.** Secondary electron (SE) images of the four alloys. 2-Hf adapted from Ref. [29].

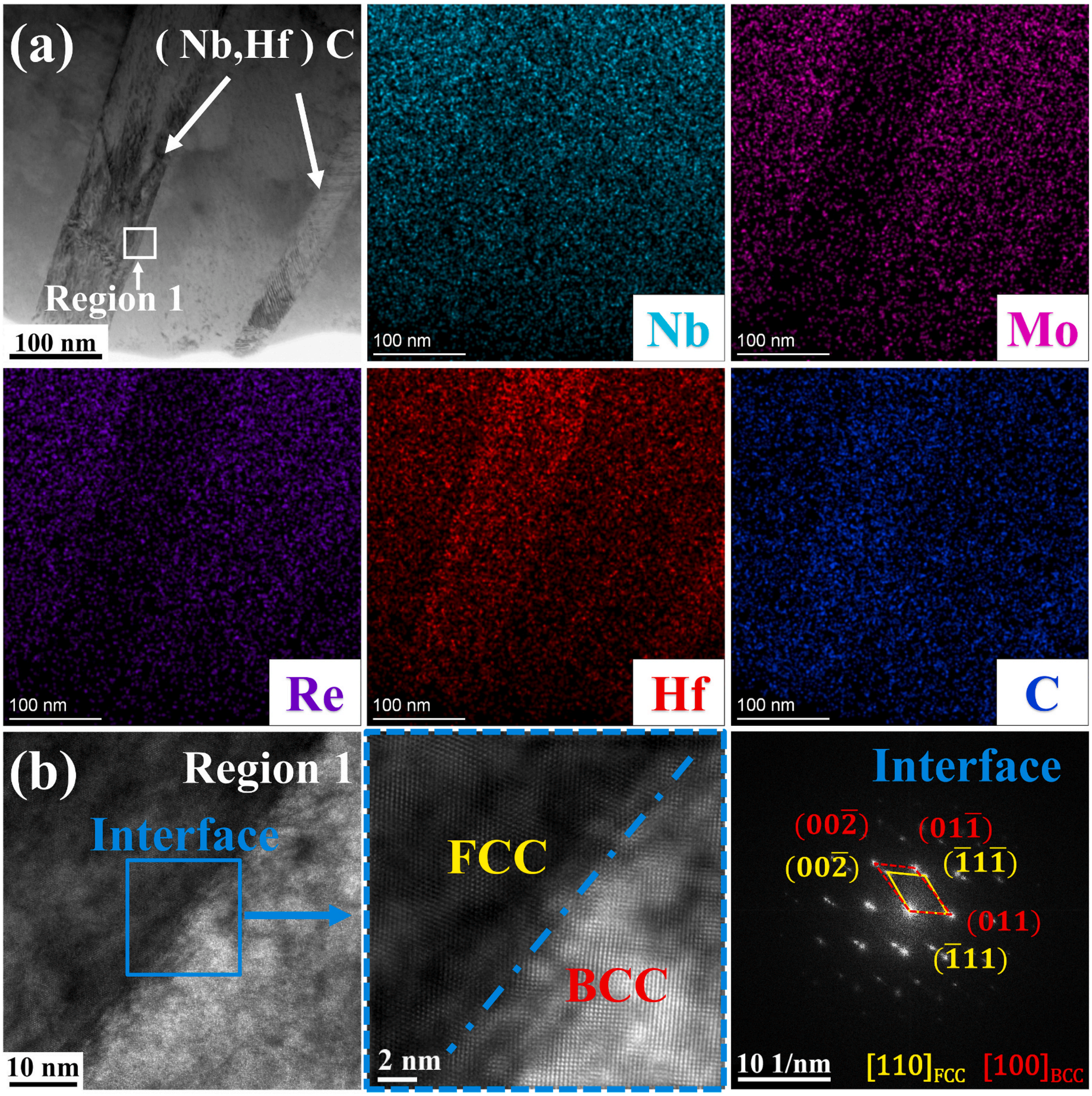


**Fig. 3.** (a) TEM bright-field (BF) image of the #4-HfRe alloy and the corresponding elemental distribution maps; (b) High-resolution TEM (HRTEM) image of Region 1, filtered HRTEM image of the Interface region, and the corresponding Fast Fourier Transform (FFT) pattern.

## 4. Discussion

### *4.1. Comparison of mechanical properties at room and elevated temperatures between the present alloys and other typical RHEAs*

The mechanical properties of the four alloys developed in this work were compared with those of representative refractory materials reported in Refs. [4,12,19,28,34–38], as summarized in Figs. 8–10. (Since most of the literature does not report experimentally measured densities, the theoretical densities calculated using the rule-of-mixtures (Eq. S(1)) were uniformly adopted to evaluate the specific yield strength when making comparisons with previously reported alloys. In the present work, the densities of the alloys were experimentally determined using the Archimedes immersion method. Both the theoretical and measured densities of the four alloys are summarized in Table S1.)

At room temperature, the specific yield strengths of #1, #2-Hf, #3-Re, and #4-HfRe are 68.90, 71.31, 80.91, and 91.16 MPa $cm^3/g$, respectively, showing no apparent advantage over the compared alloys. Nevertheless, the corresponding compressive fracture strains are >50%, >50%, 31%, and >50%, indicating exceptionally high room-temperature plastic deformability. In particular, the typical NbMoTaW alloy exhibits a room-temperature fracture strain of only 2.1%, whereas the alloys in this work reach up to 23.8 times that value. The excellent room-temperature compressive ductility implies good formability at room temperature, and suggests that components fabricated from these alloys may possess improved damage tolerance.

In addition to the outstanding room-temperature compressive ductility, the present alloys exhibit high strength at 1450 °C. The specific yield strengths at 1450 °C are 20.66, 28.37, 33.01, and 38.14 MPa $cm^3/g$ for #1, #2-Hf, #3-Re, and #4-HfRe, respectively. These values are markedly higher than those of TaNbHfZrTi, NbMoTaW, $AlMo_{0.5}NbTa_{0.5}TiZr$, and TaZrHfNbMo alloys, and are only slightly lower than those of $Re_{0.1}Hf_{0.05}Ta_{1.6}W_{0.4}(TaC)_{0.35}$ and NbMoTaWV alloys. Moreover, the high-temperature softening resistance of the alloys was assessed from the high-temperature true stress–strain curves shown in Fig. S8, and the present alloys show excellent resistance to high-temperature softening.

For NbMoTaW and NbMoTaWV, the high-temperature compressive

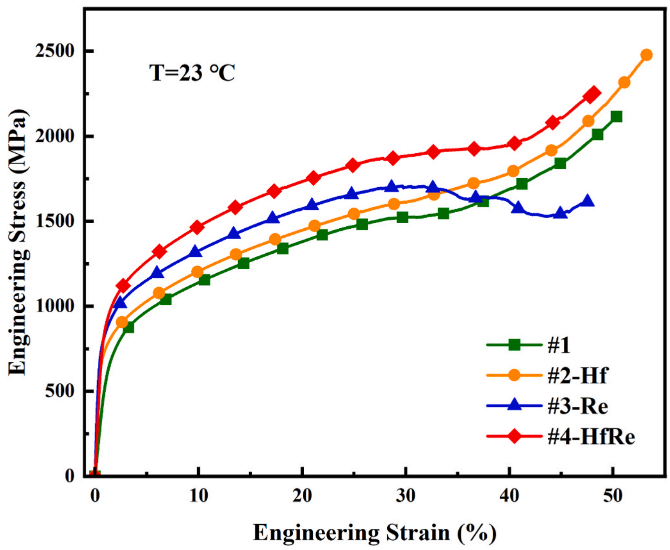


**Fig. 4.** Compressive engineering stress-strain curves of the four alloys at room temperature. 2-Hf adapted from Ref. [29].

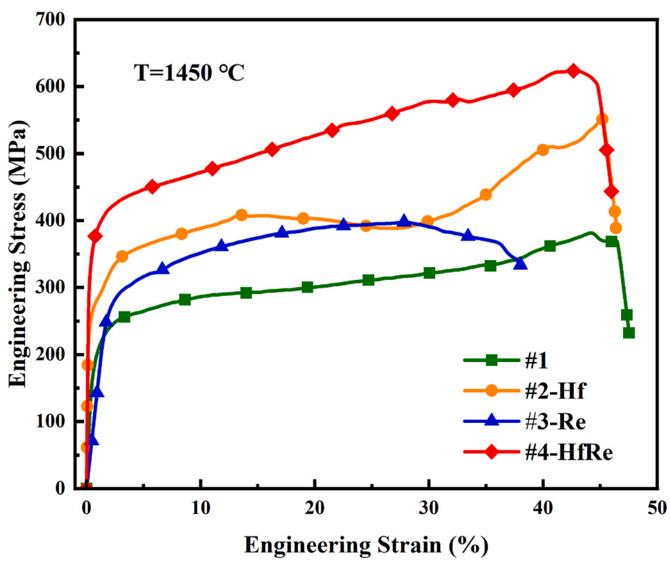


**Fig. 5.** Compressive engineering stress-strain curves of the four alloys at 1450 °C. 2-Hf adapted from Ref. [29].

yield strength is nearly identical to the ultimate compressive strength, indicating rapid softening after yielding [4]. By contrast, as summarized in Table 3, the flow stress of the present alloys decreases negligibly over the 5%–15% strain range, and decreases only slightly over the 10%–20% strain range. Specifically, the true stress drop from 10% to 20% strain is 15.00 MPa, 18.96 MPa, 2.26 MPa, and 3.96 MPa for #1, #2-Hf, #3-Re, and #4-HfRe, respectively. In comparison, the stress drops for NbMoTaW and NbMoTaWV are 177.26 MPa and 160.61 MPa, which are 44.70 and 40.50 times that of the $Re_{0.04}Hf_{0.04}Nb_{1.6}Mo_{0.3}C_{0.04}$ alloy, respectively. Moreover, because the high-temperature compressive stress–-strain curves of the reference alloys fractured prematurely and thus could not provide comparable data over a larger and identical strain range, the softening modulus was calculated over the strain interval from the peak true stress to fracture for each alloy. As summarized in Table 3, within this strain range, the four alloys developed in the present work still exhibit superior resistance to high-temperature softening. Overall, although the high-temperature strength of the present alloys at 1450 °C is slightly lower than that of typical refractory high-entropy alloys such as NbMoTaW and NbMoTaWV, their resistance to high-temperature softening is substantially superior, and they simultaneously exhibit outstanding room-temperature ductility, indicating promising application potential.

### *4.2. Influence of Hf and Re on the microstructure and mechanical properties of $Nb_{1.6}Mo_{0.3}C_{0.04}$ alloy*

All four alloys exhibit good room-temperature compressive ductility and high-temperature strength. This behavior is mainly attributed to the presence of semi-coherent carbide precipitates, which effectively enhance elevated-temperature strength without causing a pronounced degradation in room-temperature ductility.

Fig. 11 shows TEM micrographs of the #4-HfRe alloy after 40% compressive strain at room temperature. As shown, a high density of strain-induced stacking faults is generated within the carbides, and the carbides undergo pronounced shear deformation. These observations indicate that the carbides possess a certain capacity for plastic deformation at room temperature, which can accommodate and coordinate

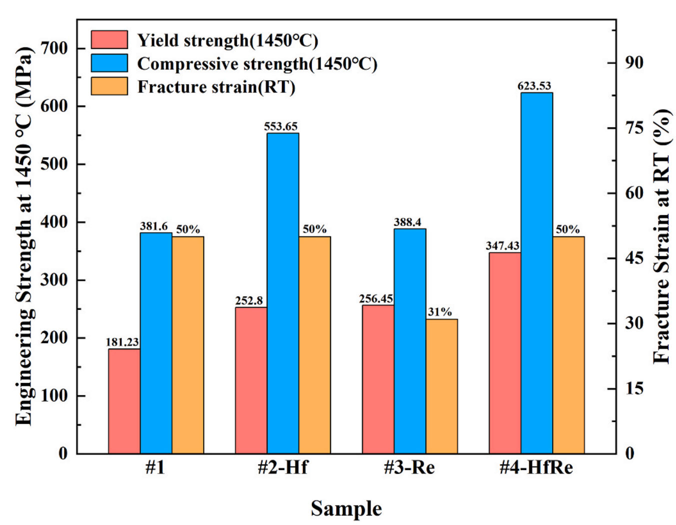


**Fig. 6.** Comparison of the compressive mechanical properties for the four alloys at room temperature and 1450 °C.

**Table 2**
Compressive mechanical properties of the four alloys at room temperature and 1450 °C.

| Alloys | Temperature (°C) | Yield strength (MPa) | Ultimate Compressive Strength (MPa) | Fracture strain (%) | Specific Yield strength ($MPa\cdot cm^3/g$) |
|---|---|---|---|---|---|
| #1 | 23 | 604.29 | >2116.60 | >50.00 | 68.90 |
| | 1450 | 181.23 | 381.60 | 44.10 | 20.66 |
| #2-Hf | 23 | 635.36 | >2477.57 | >50.00 | 71.31 |
| | 1450 | 252.80 | 553.65 | 45.21 | 28.37 |
| #3-Re | 23 | 726.54 | 1708.00 | 31.00 | 80.91 |
| | 1450 | 256.45 | 388.40 | 27.90 | 28.56 |
| #4-HfRe | 23 | 830.48 | >2254.56 | >50.00 | 91.16 |
| | 1450 | 347.43 | 623.53 | 42.30 | 38.14 |

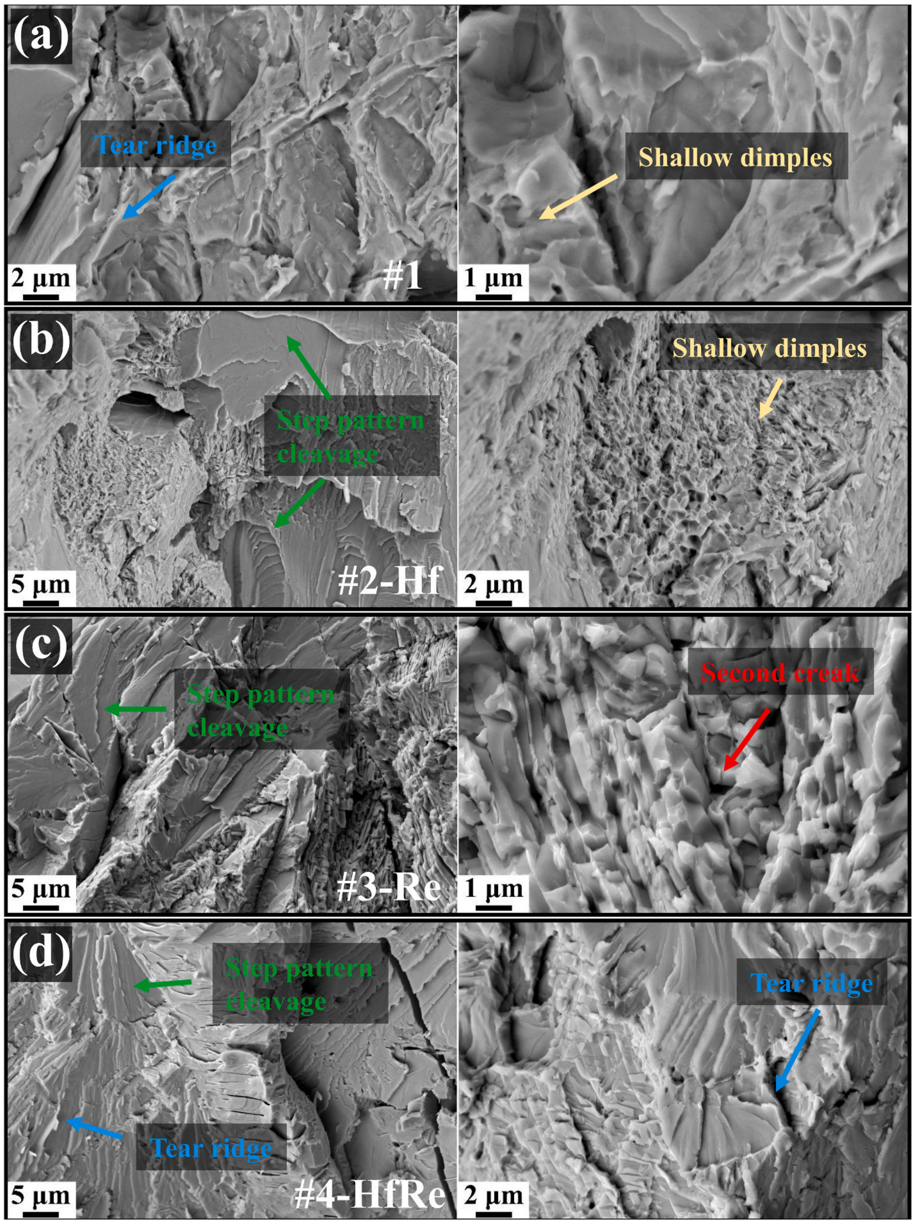


**Fig. 7.** SEM images of the fracture surfaces after room-temperature compression: (a) #1 alloy; (b) #2-Hf alloy; (c) #3-Re alloy; (d) #4-HfRe alloy. 2-Hf adapted from Ref. [29].

deformation, thereby mitigating stress/strain localization near the matrix/precipitate interfaces and reducing damage accumulation. Consequently, good room-temperature compressive ductility is retained.

The yield strength of the alloys prepared in this work can be evaluated using Eq. (1) [20].

$$\sigma_{ys} = \sigma_i + \Delta\sigma_{ss}^{sub} + \Delta\sigma_{gb} + \Delta\sigma_p \tag{1}$$

In Eq. (1), $\sigma_i$, $\Delta\sigma_{ss}^{sub}$, $\Delta\sigma_{gb}$, and $\Delta\sigma_p$ denote the intrinsic strength, solid-solution strengthening, grain-boundary strengthening, and precipitation strengthening contributions, respectively. Since the matrix compositions of the four alloys are essentially identical, the precipitation strengthening introduced by carbide precipitation is considered first. Precipitation strengthening generally proceeds through two mechanisms: (i) dislocation bypassing via the Orowan-looping mechanism and (ii) particle shearing. As evidenced by Fig. 11, the carbides undergo pronounced shear/plastic deformation during room-temperature straining; therefore, the particle-shearing mechanism should be taken into account.

Particle-shearing strengthening mainly involves three contributions: interfacial strengthening ($\Delta\sigma_{is}$), coherency strengthening ($\Delta\sigma_{cs}$), and modulus-mismatch strengthening ($\Delta\sigma_{ms}$).

$$\Delta\sigma_p = \Delta\sigma_{is} + \Delta\sigma_{cs} + \Delta\sigma_{ms} \tag{2}$$

The interfacial strengthening contribution ($\Delta\sigma_{is}$) can be calculated using the following equation [29,39–41]:

$$\Delta\sigma_{is} = \frac{0.908D_p}{t_p^2}\left(\frac{bf}{\Gamma}\right)^{\frac{1}{2}}\gamma_{APB}^{\frac{3}{2}} \tag{3}$$

$$\Gamma = \frac{Gb^2}{2\pi}\ln\sqrt{\frac{D_p^2}{2b^2f}} \tag{4}$$

The coherency strengthening contribution ($\Delta\sigma_{cs}$) can be calculated using the following equation [29,39–41]:

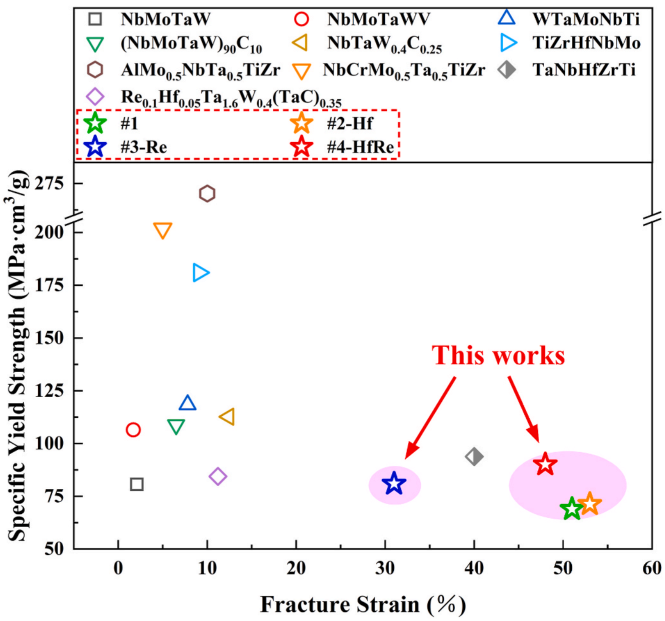


**Fig. 8.** Comparison of the specific yield strength and compressive fracture strain between the present alloys and other refractory alloys at room temperature.

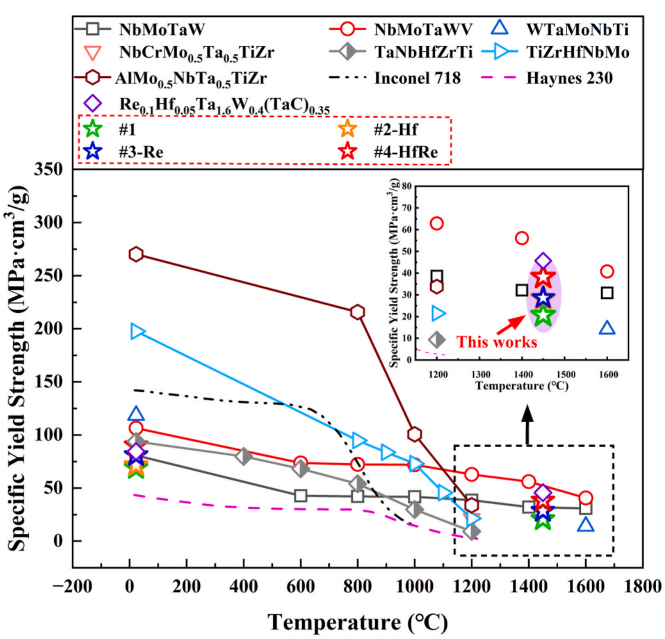


**Fig. 9.** Comparison of the specific yield strength between the present alloys and other refractory alloys at various temperatures.

$$\Delta\sigma_{cs} = 1.2 f^{\frac{1}{2}}\left[\frac{\Gamma^3 G\varepsilon}{(br)^3}\right]^{\frac{1}{4}} \tag{5}$$

$$\varepsilon = |\delta|\left[1+\frac{2G(1-2\nu_p)}{G_p(1+\nu_p)}\right] \tag{6}$$

$$\delta = \frac{a_{FCC}-\sqrt{2}a}{\sqrt{2}a} \tag{7}$$

The modulus-mismatch strengthening contribution ($\Delta\sigma_{ms}$) can be calculated using the following equation [29,39–41]:

$$\Delta\sigma_{ms} = 0.0055(\Delta G)^{\frac{3}{2}}\left(\frac{f}{\Gamma}\right)^{\frac{1}{2}} b\left(\frac{r}{b}\right)^{0.275} \tag{8}$$

$$\Delta G = |G_p - G| \tag{9}$$

In Eqs. (2)–(9), $G$ is the shear modulus of the matrix, $b$ is the Burgers vector of the matrix, $f$ is the volume fraction of precipitates, $D_p$ is the length of the platelet-like precipitates, and $t_p$ is the thickness of the platelet-like precipitates. $\gamma_{APB}$ denotes the interfacial energy between the matrix and the precipitates, $\Gamma$ is the dislocation line tension, and $G_p$ is the shear modulus of the platelet-like precipitates. $\varepsilon$ is the mismatch between dislocations and the platelet-like precipitates, $\delta$ is the lattice misfit between the precipitates and the matrix, and $\nu_p$ is the Poisson's ratio of the platelet-like precipitates. $r$ is the precipitate radius, defined as $r = 0.5D_p$. $\Delta G$ represents the shear-modulus mismatch between the matrix and the precipitates.

By substituting the parameters listed in Table 4 into Eqs. (2)–(9), the interfacial strengthening contribution is obtained as $\Delta\sigma_{is} = 19.41$ MPa. Both the coherency strengthening ($\Delta\sigma_{cs}$) and the modulus-mismatch strengthening ($\Delta\sigma_{ms}$) depend on the precipitate shear modulus $G_p$. Literature values indicate that the shear modulus of HfC can reach 180 GPa and 215.8 GPa [42]. However, as shown in Fig. 11(f and g), the precipitates exhibit pronounced stacking faults and shear deformation during room-temperature compression, suggesting an effectively lower shear modulus. Therefore, for a rough estimation of the particle-shearing strengthening, the shear modulus of the carbides was assumed to be 65, 70, and 75 GPa, respectively. The corresponding estimated strengthening contributions from the particle-shearing mechanism are summarized in Table 5. As shown in Table 5, the presence of carbides markedly enhances the alloy strength.

All four alloys are predominantly composed of a BCC matrix. Upon the addition of Hf and Re, these elements dissolve into the matrix to different extents, giving rise to varying degrees of solid-solution strengthening. Based on Ref. [47,48], a rough estimation of the solid-solution strengthening contribution was performed using a modified Labusch model. Since carbon is primarily consumed in carbide formation, its strengthening effect in the matrix was neglected in this estimation, and the increment in solid-solution strengthening was assumed to originate mainly from substitutional solute strengthening.

$$\Delta\sigma_{ss}^{sub} = A\mu\left(\sum c_i\delta_i^2\right)^{\frac{2}{3}} \tag{10}$$

$$\delta_i = \frac{9}{8}\sum c_i\left(\delta_{uij} + \beta\delta_{rij}\right) \tag{11}$$

$$\delta_{uij} = \frac{2\left(\mu_i - \mu_j\right)}{\left(\mu_i + \mu_j\right)} \tag{12}$$

$$\delta_{rij} = \frac{2(r_i - r_j)}{(r_i + r_j)} \tag{13}$$

Here, $A$ is a dimensionless constant ($A = 0.034$ [8]); $\mu$ is the shear modulus of the alloy; $c_i$ is the atomic concentration of solute element $i$; $\delta$ is the total elastic misfit; and $\beta$ is a dimensionless constant ($\beta = 8$). Using the atomic radii and shear moduli listed in Table S2, together with the semi-quantitative compositions obtained from SEM–EDS point analyses (Table S3), the solid-solution strengthening contributions of the four alloys were calculated according to Eqs. (10)–(13), and the results are summarized in Table S4. Because the SEM–EDS point analyses are semi-quantitative, these calculated values are intended only to indicate strengthening trends. As shown in Table S4, Re provides a more pronounced solid-solution strengthening effect, whereas the addition of Hf promotes carbide formation and thus yields the smallest solid-solution

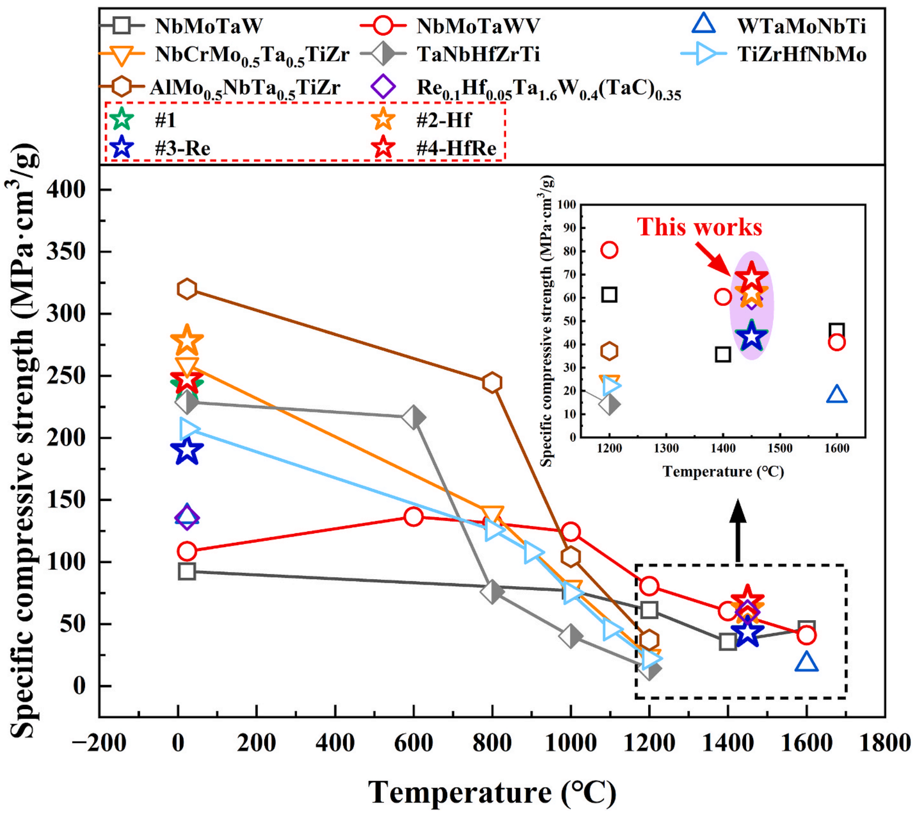


**Fig. 10.** Comparison of the specific compressive strength between the present alloys and other refractory alloys at various temperatures.

**Table 3**
Comparison of the high-temperature compressive softening modulus between the four prepared alloys and other typical refractory alloys.

| Alloys | T (°C) | Flow softening modulus (MPa) | | | Reference |
|---|---|---|---|---|---|
| | | 5–15% | 10–20% | Peak true stress to fracture | |
| #1 | 1450 | 2.80 | −150.00 | −118.22 | This work |
| #2-Hf | 1450 | 79.20 | −189.60 | −109.51 | This work |
| #3-Re | 1450 | 179.70 | −22.60 | −322.14 | This work |
| #4-HfRe | 1450 | 7.10 | −39.60 | −189.95 | This work |
| NbMoTaW | 1200 | −213.90 | −965.20 | −989.10 | [4] |
| NbMoTaWV | 1400 | −3320 | −1772.60 | −2779.48 | [4] |
| $Re_{0.1}Hf_{0.05}Ta_{1.6}W_{0.4}(TaC)_{0.35}$ | 1450 | 364.20 | −1606.10 | −2145.25 | [19] |

strengthening contribution to the matrix in the #2-Hf alloy.

In addition, the grain-boundary strengthening contribution was evaluated using the Hall–Petch relationship [49].

$$\Delta\sigma_{gb} = kd^{-0.5} \tag{14}$$

Here, $\Delta\sigma_{gb}$ denotes the grain-boundary strengthening contribution; $k$ is the Hall–Petch coefficient, which is taken as 340 MPa $\mu m^{-0.5}$ for Nb-based alloys according to Ref. [8]; and $d$ is the average grain size. The grain-boundary strengthening contributions were semi-quantitatively estimated using Eq. (14), and the results are summarized in Table S4. As indicated in Table S4, although the addition of Hf and Re leads to grain refinement, the corresponding grain-boundary strengthening contributions are relatively small and do not play a dominant role in the overall strengthening of the alloys.

In the present work, minor additions of Re and Hf exert a pronounced influence on the mechanical properties of the #1 alloy, mainly because even trace additions can significantly alter the carbide precipitation behavior. Table 6 and Fig. S6 summarize the carbide volume fraction, which was quantified by Rietveld full-pattern refinement using the GSAS software, and the matrix/precipitate misfit, calculated via Eq. (7) based on the lattice parameters in Table 1, for the four alloys. For #1 alloy, the carbide volume fraction is 2.45%. With a minor Re addition, the amount of carbide precipitates decreases markedly to 2.07%, corresponding to a reduction of 15.50%. This decrease is attributed to the ability of Re to significantly increase the carbon solubility in the alloy [50]. A reduced carbide fraction is expected to adversely affect strengthening. Nevertheless, Re provides strong solid-solution strengthening and can increase the recrystallization temperature of the BCC matrix, which benefits both room-temperature and high-temperature strength [24–27]. Therefore, despite the reduction in carbide fraction, the high-temperature strength is still improved after Re addition. It should be noted that, although Re has a relatively small atomic radius and typically decreases the lattice parameter of the BCC phase, the increased carbon solubility offsets this effect; consequently, the lattice parameter of the #3-Re alloy shows no obvious change, and the matrix/precipitate misfit remains nearly unchanged. After strengthening, both the room-temperature compressive ductility and the resistance to high-temperature softening decrease to some extent.

With the minor addition of Hf alone, the carbide content increases significantly because Hf is a strong carbide-forming element. The carbide volume fraction reaches 2.86%, representing increases of 16.70%

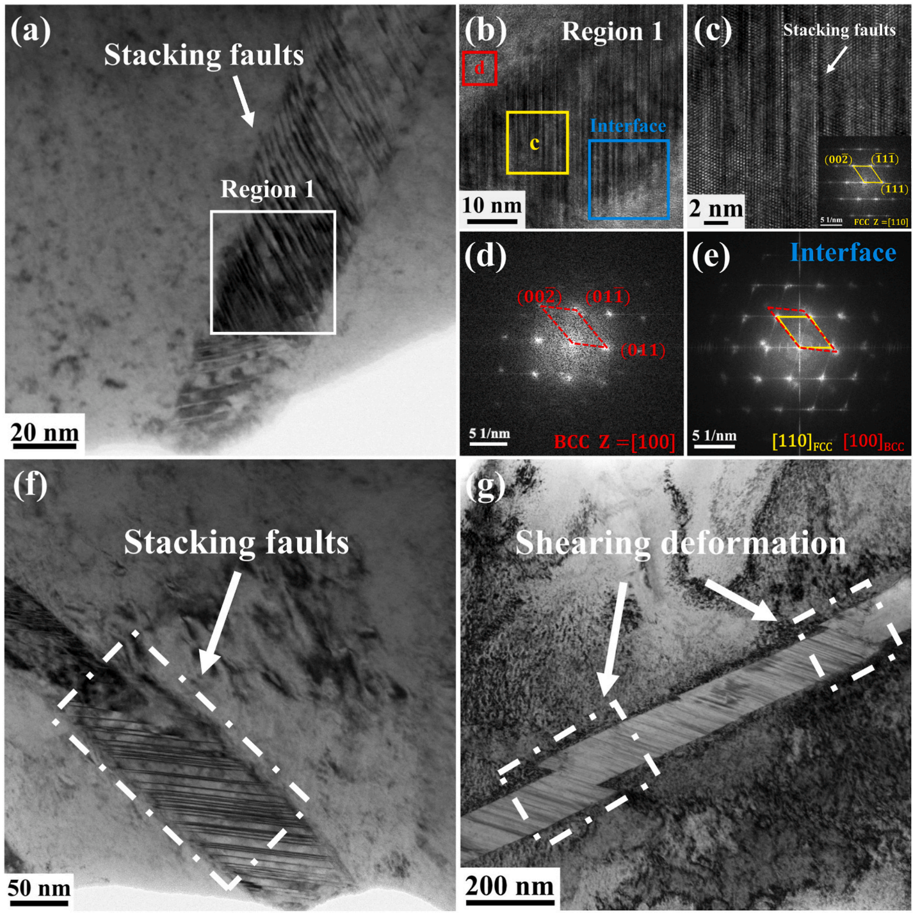


**Fig. 11.** TEM images of the #4-HfRe alloy after 40% room-temperature compressive strain: (a) Deformation-induced stacking faults within a carbide; (b) High-resolution TEM (HRTEM) image of Region 1; (c) HRTEM image of area 'c' in (b) and the corresponding Fast Fourier Transform (FFT) pattern; (d, e) Indexed FFT patterns from area 'd' and the Interface region in (b), respectively; (f, g) Strain-induced stacking faults and shear-deformed carbides.

**Table 4**
Parameters used in the calculations of Eqs. (2)-(9).

| Parameter | Description | Value | Reference |
|---|---|---|---|
| $G$ | Shear modulus of matrix | 53.70 (GPa) | [43–45] |
| $G_p$ | Shear modulus of precipitates | 65/70/75 (GPa) | This work |
| $b$ | Burgers vector of matrix | 0.2842 (nm) | This work |
| $D_p$ | Diameter of the precipitates | 3585 (nm) | This work |
| $r$ | $r$ is the radius of lamellar precipates, $r = 0.5D_p$ | 1779.25 (nm) | This work |
| $t_p$ | Thickness of the precipitates | 65.01 (nm) | This work |
| $f$ | the volume fraction of precipitated phase | 2.83% | This work |
| $\gamma_{APB}$ | Interracial energy between MC precipates and matrix phases | 0.84 (J/m$^2$) | [46] |
| $\nu_p$ | Poisson's ratio of precipates | 0.22 | [42] |
| $\delta$ | the lattice misfits between the plate-like precipitates and the matrix | 3.84% | This work |

**Table 5**
Calculated strengthening contributions for the particle shearing mechanism under assumed precipitate shear modulus values.

| $G_p$ (GPa) | $\Delta\sigma_{is}$ (MPa) | $\Delta\sigma_{cs}$ (MPa) | $\Delta\sigma_{ms}$ (MPa) | $\Delta\sigma_p$ (MPa) |
|---|---|---|---|---|
| 65 | 19.41 | 55.96 | 40.44 | 115.81 |
| 70 | 19.41 | 55.53 | 70.06 | 145.00 |
| 75 | 19.41 | 55.14 | 104.65 | 179.2 |

and 38.20% compared with #1 and #3-Re, respectively. The increased carbide fraction contributes positively to alloy strengthening. Moreover, the carbides transform from NbC to a complex carbide (Hf, Nb)C, which exhibits improved high-temperature stability [51,52] and is beneficial for high-temperature strength. As summarized in Tables 1 and 5, differences are observed in the lattice parameters of the BCC matrix as well as those of the MC-type carbides (NbC and (Hf, Nb)C) among the four alloys. Notably, although the addition of Hf leads to an increase in the lattice parameter of the BCC matrix, the lattice parameter of the (Hf, Nb)C carbide also increases. As a result, the lattice misfit between the carbide and the BCC matrix is reduced from 5.54% in the #1 alloy to 3.68% in the #2-Hf alloy, indicating an improved lattice matching between the two phases. Notably, although the addition of Hf enhances carbide strengthening, the improvement in matrix strength is limited due to the relatively small Hf content, as indicated by the solid-solution strengthening contribution discussed above. Consequently, the #2-Hf alloy exhibits slightly reduced flow-stress stability during high-temperature compression.

**Table 6**
Volume fraction of carbides and their lattice misfit with the matrix in the four alloys.

| Alloys | Types of carbide | Carbide volume fraction (%) | Misfit $\delta$ (%) |
|---|---|---|---|
| #1 | NbC | 2.45 | 5.54 |
| #2-Hf | (Hf, Nb)C | 2.86 | 3.68 |
| #3-Re | NbC | 2.07 | 5.52 |
| #4-HfRe | (Hf, Nb)C | 2.83 | 3.84 |

Fig. S10 shows the optical micrograph of the #4-HfRe alloy after compression at 1450 °C. The platelet-like carbides remain uniformly distributed within the grains and partially along grain boundaries, where they continue to act as stable obstacles for dislocation pinning. This microstructural feature is consistent with that reported in the authors' previous work on the $Zr_{0.04}Nb_{1.6}Mo_{0.3}C_{0.04}$ alloy subjected to compression at 1450 °C. In addition, the MC carbides contain a high density of dislocations and abundant stacking-fault bands. These characteristics indicate that the carbides have exceeded their brittle-to-ductile transition temperature, and multiple dislocation slip systems are thermally activated. Consequently, the carbides deform through dislocation slip and stacking-fault formation, thereby actively accommodating local plastic strain [29].

Accordingly, the introduction of Hf promotes the transformation of NbC into (Hf, Nb)C carbides, while simultaneously regulating carbide morphology and spatial distribution and reducing the interfacial lattice misfit with the BCC matrix. Moreover, the higher melting point and improved thermal stability of Hf-containing carbides help maintain their structural integrity during high-temperature deformation, thereby preserving their strengthening effectiveness. Owing to their high elastic modulus and high melting point, MC-type carbides effectively impede dislocation motion and share the applied load, leading to enhanced high-temperature strength and improved resistance to thermal softening. Furthermore, the semi-coherent interfaces with reduced lattice misfit between the carbides and the BCC matrix enhance interfacial dislocation pinning, which suppresses dynamic recrystallization at elevated temperatures and enables the alloy to maintain relatively stable flow stress over a wide strain range [18,29].

When Hf and Re are added simultaneously, Hf, as a strong carbide-forming element, counteracts the Re-induced suppression of carbide precipitation. Accordingly, the total carbide fraction is comparable to that of the Hf-containing alloy, while the matrix/precipitate misfit remains at a low level of 3.84%. Re plays a critical role in strengthening the BCC matrix, whereas Hf enables effective carbide strengthening. Owing to the synergistic effects of Re and Hf, the alloy achieves the best high-temperature mechanical performance, while maintaining good room-temperature compressive ductility, together with outstanding high-temperature strength and resistance to high-temperature softening. The above discussion indicates that the Re–Hf synergy is a promising approach to markedly improve the overall mechanical performance of carbide-strengthened refractory alloys.

## 5. Conclusions

Four carbide-strengthened Nb–Mo-based alloys with nominal compositions of $Nb_{1.6}Mo_{0.3}C_{0.04}$, $Nb_{1.6}Mo_{0.3}C_{0.04}Hf_{0.04}$, $Nb_{1.6}Mo_{0.3}C_{0.04}Re_{0.04}$, and $Nb_{1.6}Mo_{0.3}C_{0.04}Hf_{0.04}Re_{0.04}$ were fabricated by vacuum arc melting. Their microstructures and mechanical properties were investigated, with particular emphasis on the effects of Re and Hf on the room-temperature and high-temperature mechanical behavior of the $Nb_{1.6}Mo_{0.3}C_{0.04}$ alloy. The main conclusions are as follows.

(1) The carbide-strengthened $Nb_{1.6}Mo_{0.3}$-based alloys exhibit good room-temperature compressive ductility, high-temperature strength, and excellent resistance to high-temperature softening. In particular, the $Nb_{1.6}Mo_{0.3}C_{0.04}Hf_{0.04}Re_{0.04}$ alloy shows a room-temperature compressive fracture strain of >50%, which is 23.8 times that of the NbMoTaW alloy. At 1450 °C, its specific yield strength reaches 38.14 MPa $cm^3$/g, which is 1.19 times that of the NbMoTaW alloy at 1400 °C. Moreover, over the true strain range of 10%–20% at 1450 °C, the true-stress drop of NbMoTaW is 177.26 MPa, whereas that of $Re_{0.04}Hf_{0.04}Nb_{1.6}Mo_{0.3}C_{0.04}$ is only 3.96 MPa; thus, the stress drop of NbMoTaW is 44.70 times that of $Nb_{1.6}Mo_{0.3}C_{0.04}Hf_{0.04}Re_{0.04}$.

(2) Carbides play a critical role in achieving both good room-temperature ductility and outstanding high-temperature strength. The carbides formed in this work are plastically deformable at room temperature via shear mechanism, which effectively alleviates stress concentration during deformation. The enhancement in high-temperature strength and resistance to thermal softening can be attributed to the dual roles of the carbides: load sharing arising from their high elastic modulus, and dislocation pinning enabled by the low-misfit semi-coherent interfaces with the BCC matrix. In addition, the carbides accommodate plastic strain through dislocation slip and stacking-fault formation, which not only impedes dislocation motion and suppresses dynamic recrystallization but also effectively avoids stress concentration. Owing to the synergistic action of these mechanisms, the alloys are able to maintain high flow stress and excellent resistance to high-temperature softening even under elevated temperatures and large strains.

(3) Minor additions of Hf and Re significantly modify the carbide precipitation behavior in $Nb_{1.6}Mo_{0.3}C_{0.04}$, thereby affecting both room-temperature and high-temperature mechanical responses. Re provides pronounced strengthening of the BCC matrix, while Hf promotes carbide precipitation, reduces carbon supersaturation in the matrix, and counteracts the Re-induced suppression of carbide formation, leading to the precipitation of composite carbides with enhanced coherency with the matrix. Through the synergistic effects of Hf and Re, the alloy achieves a desirable combination of good room-temperature ductility and exceptionally high high-temperature strength. This design strategy provides useful guidance for further improving the mechanical performance of carbide-strengthened Nb–Mo-based refractory alloys by balancing high-temperature strength retention and room-temperature processability for ultrahigh-temperature load-bearing applications.

## CRediT authorship contribution statement

**Y.J. Hou:** Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **J.X. Fang:** Writing – review & editing, Supervision, Resources, Investigation, Funding acquisition, Conceptualization. **Q. Su:** Data curation, Investigation, Visualization, Resources. **H.T. He:** Visualization, Supervision, Methodology, Investigation. **X. Mei:** Supervision, Methodology, Investigation. **G.F. Long:** Investigation, Data curation. **L.Y. Hao:** Supervision, Investigation. **M. Wen:** Visualization, Supervision, Investigation, Funding acquisition.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgements

The works is supported by the General Program of National Natural Science Foundation of China (52475193), Key Project of Natural Science Foundation of Guizhou ([2021]053), Yunnan Fundamental Research Projects (grant NO. 202201BE070001-059), Yunnan Xingdian Yingcai Support Program Young Talents Special Support Project (KKXX202401019), Scientific and Technological Project of Yunnan Precious Metals Laboratory (YPML-20240502017) and Yunnan Fundamental Research Projects (grant NO. 202301AT070342). The authors gratefully acknowledge Dr. Q. Su from CNPC Offshore Engineering Company Limited for providing part of the experimental equipment used in this work.

## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.

org/10.1016/j.msea.2026.150047.

## Data availability

Data will be made available on request.

## References


[1] P. Tsakiropoulos, On the design of ultra-high-temperature material systems, Materials 18 (2025) 5203, https://doi.org/10.3390/ma18225203.
[2] A. Epishin, B. Fedelich, M. Finn, G. Künecke, B. Rehmer, G. Nolze, C. Leistner, N. Petrushin, I. Svetlov, Investigation of elastic properties of the single-crystal nickel-base superalloy CMSX-4 in the temperature interval between room temperature and 1300 °C, Crystals 11 (2021) 152, https://doi.org/10.3390/cryst11020152.
[3] M. Mostafaei, S.M. Abbasi, Influence of Zr content on the incipient melting behavior and stress-rupture life of CM247 LC nickel base superalloy, J. Alloys Compd. 648 (2015) 1031–1037, https://doi.org/10.1016/j.jallcom.2015.07.104.
[4] O.N. Senkov, G.B. Wilks, J.M. Scott, D.B. Miracle, Mechanical properties of Nb25Mo25Ta25W25 and V20Nb20Mo20Ta20W20 refractory high entropy alloys, Intermetallics 19 (2011) 698–706, https://doi.org/10.1016/j.intermet.2011.01.004.
[5] L.S. Bowling, A.T. Wang, N.R. Philips, W.T. Riffe, D.E. Matejczyk, J.M. Skelton, P. E. Hopkins, J.M. Fitz-Gerald, S.R. Agnew, Thermophysical modeling of niobium alloys informs materials selection and design for high-temperature applications, Mater. Des. 248 (2024) 113456, https://doi.org/10.1016/j.matdes.2024.113456.
[6] M. Fujikura, A. Kasama, R. Tanaka, S. Hanada, Effect of alloy chemistry on the high temperature strengths and room temperature fracture toughness of advanced Nb-based alloys, Mater. Trans. 45 (2004) 493–501, https://doi.org/10.2320/matertrans.45.493.
[7] D.L. Anton, D.B. Snow, L.H. Favrow, A.F. Giamei, Dispersion Strengthening of High Temperature Niobium Alloys, United Technologies Research Center, East Hartford, CT, USA, 1989, Report No. R89-917437-3. https://www.researchgate.net/publication/235090650_Dispersion_Strengthening_of_High_Temperature_Niobium_Alloys.
[8] O.N. Senkov, S.I. Rao, T.M. Butler, T.I. Daboiku, K.J. Chaput, Microstructure and properties of nb-mo-zr based refractory alloys, Int. J. Refract. Metals Hard Mater. 92 (2020) 105321, https://doi.org/10.1016/j.ijrmhm.2020.105321.
[9] O.N. Senkov, S.I. Rao, T.M. Butler, K.J. Chaput, Ductile Nb alloys with reduced density and cost, J. Alloys Compd. 808 (2019) 151685, https://doi.org/10.1016/j.jallcom.2019.151685.
[10] E. Panina, N. Yurchenko, A. Tojibaev, M. Mishunin, S. Zherebtsova, N. Stepanova, Mechanical properties of (HfCo)100-x(NbMo)x refractory high-entropy alloys with a dual-phase bcc-B2 structure, Mater. Sci. Eng., A 927 (2022) 167013, https://doi.org/10.1016/j.jallcom.2022.167013.
[11] N.N. Guo, L. Wang, L.S. Luo, X.Z. Li, R.R. Chen, Y.Q. Su, J.J. Guo, H.Z. Fu, Effect of composing element on microstructure and mechanical properties in MoNbHfZrTi multi-principle component alloys, Mater. Sci. Eng., A 69 (2016) 13–20, https://doi.org/10.1016/j.intermet.2015.10.011.
[12] Y. Wan, J. Mo, X. Wang, Z. Zhang, B. Shen, X. Liang, Mechanical properties and phase stability of WTaMoNbTi refractory high-entropy alloy at elevated temperatures, Acta Metall. Sin. 34 (2021) 1585–1590, https://doi.org/10.1007/s40195-021-01263-9.
[13] J. Xu, M. Qin, S. Du, P. Kumar, J. Zhu, Y. Jia, Z. Yan, Y. Xie, P. Dong, K. Feng, Z. Li, Q. Zhou, X. Liang, U. Ramamurty, Enhancing the strength and plasticity of laser powder bed fused NbMoTaW refractory high-entropy alloy via Ti alloying, Mater. Sci. Eng., A 1001 (2024) 175043, https://doi.org/10.1016/j.jallcom.2024.175043.
[14] Q. Wang, Z. Liu, Y. Wang, X. Wang, The effect of (ZrB2-SiC) addition on microstructure and mechanical properties of NbMo-matrix composites fabricated by hot-pressing, Int. J. Refract. Metals Hard Mater. 87 (2020) 105098, https://doi.org/10.1016/j.ijrmhm.2019.105098.
[15] Q. Wang, Z. Liu, Y. Wang, Y. Sun, Y. Gong, High temperature compression properties of hot-pressed NbMo-matrix composites reinforced with ZrB2 particles, Int. J. Refract. Metals Hard Mater. 77 (2018) 132–140, https://doi.org/10.1016/j.ijrmhm.2018.08.007.
[16] Z. Liu, Q. Wang, Y. Gao, Y. Wang, Y. Sun, Y. Gong, Preparation and properties of hot-pressed NbMo-matrix composites reinforced with ZrB2 particles, Int. J. Refract. Metals Hard Mater. 68 (2017) 104–112, https://doi.org/10.1016/j.ijrmhm.2017.06.011.
[17] N.N. Guo, L. Wang, L.S. Luo, X.Z. Li, R.R. Chen, Y.Q. Su, J.J. Guo, H.Z. Fu, Microstructure and mechanical properties of in-situ MC-carbide particulates-reinforced refractory high-entropy Mo0.5NbHf0.5ZrTi matrix alloy composite, Intermetallics 69 (2016) 74–77, https://doi.org/10.1016/j.intermet.2015.09.011.
[18] Q. Wei, X. Xu, Q. Shen, G. Luo, J. Zhang, J. Li, Q. Fang, C.-T. Liu, M. Chen, T.-G. Nieh, J. Chen, Metal-carbide eutectics with multiprincipal elements make superrefractory alloys, Sci. Adv. 8 (27) (2022), https://doi.org/10.1126/sciadv.abo2068 eabo2068.
[19] H. He, J. Fang, Z. Yang, T. Sun, B. Ma, H. Chen, T. Guo, W. Wang, Y. Shen, M. Wen, Effects of Hf and C on microstructure and mechanical properties of Re0.1HfxTa1.6W0.4(TaC)y refractory medium-entropy alloy, Int. J. Refract. Metals Hard Mater. 121 (2024) 106668, https://doi.org/10.1016/j.ijrmhm.2024.106668.
[20] H.T. He, J.X. Fang, Z. Yang, T. Sun, B. Ma, H.T. Chen, T.T. Guo, W.B. Wang, Y. J. Wang, Ductile Re0.1Ta1.9W0.2Cx refractory alloys with excellent elevated-temperature strength, Mater. Sci. Eng., A 915 (2024) 147217, https://doi.org/10.1016/j.msea.2024.147217.
[21] J. Wang, F. Lu, W. Li, P. Hu, S. Lu, C. Xiao, Effect of Hf addition on the microstructure and tensile properties at medium temperature of a cast superalloy K4222, Mater. Sci. Eng., A 942 (2025) 148680, https://doi.org/10.1016/j.msea.2025.148680.
[22] S. Zhang, X. Guo, Microstructure, mechanical properties and oxidation resistance of Nb silicide based ultrahigh temperature alloys with Hf addition, Mater. Sci. Eng., A 645 (2015) 88–98, https://doi.org/10.1016/j.msea.2015.08.006.
[23] H.T. He, J.X. Fang, J.X. Wang, T. Sun, Z. Yang, B. Ma, H.T. Chen, M. Wen, Carbide-reinforced Re0.1Hf0.25NbTaW0.4 refractory high-entropy alloy with excellent room and elevated temperature mechanical properties, Int. J. Refract. Metals Hard Mater. 116 (2023) 106349, https://doi.org/10.1016/j.ijrmhm.2023.106349.
[24] B. Wei, Y. Lin, Z. Huang, L. Huang, K. Zhou, L. Zhang, L. Zhang, A novel Re-free Ni-based single-crystal superalloy with enhanced creep resistance and microstructure stability, Acta Mater. 240 (2022) 118336, https://doi.org/10.1016/j.actamat.2022.118336.
[25] J. Zhang, Y. Hu, Q. Wei, Y. Xiao, P. Chen, G. Luo, Q. Shen, Microstructure and mechanical properties of RexNbMoTaW high-entropy alloys prepared by arc melting using metal powders, J. Alloys Compd. 827 (2020) 154301, https://doi.org/10.1016/j.jallcom.2020.154301.
[26] J. Yang, S. Wang, K. Wang, Y. Fang, H. Wang, L. Wang, H. Xing, P. Hu, Molybdenum-Rhenium alloy: a focused review of strengthening-toughening mechanism and method, Int. J. Refract. Metals Hard Mater. 134 (2026) 107408, https://doi.org/10.1016/j.ijrmhm.2025.107408.
[27] H. Xu, L. Huang, W. Zhang, J. Liang, X. Lin, X. Zhang, X. Gao, J. Li, Mechanical and microstructural responses in molybdenum-rhenium alloys under hot compressions, J. Mater. Res. Technol. 30 (2024) 3877–3885, https://doi.org/10.1016/j.jmrt.2024.04.063.
[28] H. He, J. Wang, Y. Cao, X. Yang, G. Zhao, W. Yang, J. Fang, Effect of Re and C on mechanical properties of NbTaW0.4 refractory medium-entropy alloy at elevated temperature, J. Alloys Compd. 931 (2023) 167421, https://doi.org/10.1016/j.jallcom.2022.167421.
[29] H. He, Z. Xiao, X. Mei, L. Hao, Y. Hou, G. Long, X. Zhou, T. Guo, W. Wang, T. Fu, J. Wei, J. Fang, Y. Wang, The effect of Zr, Ti, and Hf on the microstructure and mechanical properties of the Nb1.6Mo0.3C0.04 refractory alloy, J. Alloys Compd. 1053 (2026) 186192, https://doi.org/10.1016/j.jallcom.2026.186192.
[30] L. Yao, L. Wang, X. Song, R. Cui, B. Li, Q. Lv, L. Luo, Y. Su, J. Guo, H. Fu, Microstructure evolution and toughening mechanism of a Nb-18Si-5HfC eutectic alloy created by selective laser melting, J]. Materials 15 (3) (2022) 1190, https://doi.org/10.3390/ma15031190.
[31] Z.-G. Yang, M. Enomoto, Calculation of the interfacial energy of B1-type carbides and nitrides with austenite, Metall. Mater. Trans. 32A (2) (2001) 267–274, https://doi.org/10.1007/s11661-001-0258-8.
[32] W.E. Luecke, L. Ma, S.M. Graham, M.A. Adler, Repeatability and Reproducibility of Compression Strength Measurements Conducted According to ASTM E9, 1679, National Institute of Standards and Technology Technical Note, 2009, pp. 1–40, https://doi.org/10.6028/NIST.TN.1679.
[33] M.S. Loveday, T. Gray, J. Aegerter, Tensile Testing of Metallic Materials: a Review, National Physical Laboratory Technical Report, 2004, pp. 1–167. GRD1-2000-25021, https://www.npl.co.uk/products-services/advanced-materials/Tensile-testing.
[34] Y. Wan, J. Mo, X. Wang, Z. Zhang, B. Shen, X. Liang, Mechanical properties and phase stability of WTaMoNbTi Refractory high-entropy alloy at elevated temperatures, Acta Metall. Sin. Engl. Lett. 34 (2021) 1585–1590, https://doi.org/10.1007/s40195-021-01263-9.
[35] Y. Wan, X. Wang, Z. Zhang, J. Mo, B. Shen, X. Liang, Structures and properties of the (NbMoTaW)100-xCx high-entropy composites, J. Alloys Compd. 889 (2021) 161645, https://doi.org/10.1016/j.jallcom.2021.161645.
[36] N.N. Guo, L. Wang, L.S. Luo, X.Z. Li, Y.Q. Su, J.J. Guo, H.Z. Fu, Microstructure and mechanical properties of refractory MoNbHfZrTi high-entropy alloy, Mater. Des. 81 (2015) 87–94, https://doi.org/10.1016/j.matdes.2015.05.019.
[37] O.N. Senkov, S.V. Senkova, C. Woodward, Effect of aluminum on the microstructure and properties of two refractory high-entropy alloys, Acta Mater. 68 (2014) 214–228, https://doi.org/10.1016/j.actamat.2014.01.029.
[38] O.N. Senkov, C.F. Woodward, Microstructure and properties of a refractory NbCrMo0.5Ta0.5TiZr alloy, Mater. Sci. Eng., A 529 (2011) 311–320, https://doi.org/10.1016/j.msea.2011.09.033.
[39] Y. Tang, Y. Kang, L. Yue, X. Jiao, Mechanical properties optimization of a Cu-Be-Co-Ni alloy by precipitation design, J. Alloys Compd. 695 (2017) 613–625, https://doi.org/10.1016/j.jallcom.2016.11.014.
[40] J.F. Nie, B.C. Muddle, Microstructural design of high-strength aluminum alloys, JPE 19 (1998) 543–551, https://doi.org/10.1361/105497198770341734.
[41] V. Gerold, H.J. Gudladt, J. Lendvai, Microstructure and deformation behaviour of Age hardenable Al-Li single crystals, Phys. Status Solidi 131 (2) (1992) 509–522, https://doi.org/10.1002/pssa.2211310208.
[42] R. He, L. Fang, T. Han, G. Yang, G. Ma, J. Liu, X. Chen, L. Xie, L. Liu, Q. Li, Y. Tang, H. Liang, Y. Zou, F. Peng, Elasticity, mechanical and thermal properties of polycrystalline hafnium carbide and tantalum carbide at high pressure, J. Eur. Ceram. Soc. 42 (2022) 5220–5228, https://doi.org/10.1016/j.jeurceramsoc.2022.06.039.
[43] J.M. Dickinson, P.E. Armstrong, Temperature dependence of the elastic constants of molybdenum, J. Appl. Phys. 38 (2) (1967) 602–606, https://doi.org/10.1063/1.1709381.
[44] K.J. Carroll, Elastic Constants of Niobium from 4.2° to 300°K, J. Appl. Phys. 36 (11) (1965) 3689–3690, https://doi.org/10.1063/1.1703072.

[45] M.L. Frey, J.E. Lonnee, G.W. Shannette, Elastic constants of niobium-zirconium, hafnium, and tungsten alloys, J. Appl. Phys. 49 (8) (1978) 4406–4410, https://doi.org/10.1063/1.325493.
[46] N. Zhao, X. Ma, J. Wang, L. Zhong, H. Deng, First-principles study of NbC/Nb interface stability and electronic structure, Mater. Chem. Phys. 328 (2024) 129861, https://doi.org/10.1016/j.matchemphys.2024.129861.
[47] O.N. Senkov, J.M. Scott, S.V. Senkova, D.B. Miracle, C.F. Woodward, Microstructure and room temperature properties of a high-entropy TaNbHfZrTi alloy, J. Alloys Compd. 509 (2011) 6043–6048, https://doi.org/10.1016/j.jallcom.2011.02.171.
[48] H.W. Yao, J.W. Qiao, J.A. Hawk, H.F. Zhou, M.W. Chen, M.C. Gao, Mechanical properties of refractory high-entropy alloys: experiments and modeling, J. Alloys Compd. 696 (2017) 1139–1150, https://doi.org/10.1016/j.jallcom.2016.11.188.
[49] E.O. Hall, The deformation and ageing of mild steel: III Discussion of results, Proc. Phys. Soc. B 64 (1951) 747, https://doi.org/10.1088/0370-1301/64/9/303.
[50] H.M. Rosenberg, H. Nowotny, A survey of the rhenium-carbon system, Mater. Sci. Eng., A 2 (1967) 297–304, https://doi.org/10.1016/0025-5416(67)90037-6.
[51] D. Lang, L. Karge, J. Schatte, R. Gilles, R. Dallinger, I. Weißensteiner, P. Staron, W. Knabl, S. Primig, H. Clemens, Evolution of strain-induced hafnium carbides in a molybdenum base Mo-Hf-C alloy studied by small-angle neutron scattering and complementary methods, J. Alloys Compd. 688 (2016) 619–631, https://doi.org/10.1016/j.jallcom.2016.06.282.
[52] Y. Fu, Y. Zhang, H. Chen, X. Yin, J. Zhang, J. Sun, Q. Fu, Ultra-high temperature resistance of one-dimensional hafnium carbide wrapped with pyrolytic carbon up to 2450 °C, Corros. Sci. 195 (2022) 110015, https://doi.org/10.1016/j.corsci.2021.110015.